

NTIRE 2026 Mobile Real-World Image Super-Resolution Challenge Factsheet

- Representation-Aligned Adversarial Diffusion Compression for Perceptual Real-World Super-Resolution-

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1. Introduction

This factsheet template is meant to structure the description of the contributions made by each participating team in the NTIRE 2026 challenge on mobile real-world image super-resolution.

Ideally, all the aspects enumerated below should be addressed. The provided information, the codes/executables, and the achieved performance on the testing data are used to decide the awardees of the NTIRE 2026 challenge.

Reproducibility is a must and needs to be checked for the final test results in order to qualify for the NTIRE awards.

The main winners will be decided based on overall performance and a number of awards will go to novel, interesting solutions and to solutions that stand up as the best in a particular subcategory the judging committee will decide. Please check the competition webpage and forums for more details.

The winners, the awardees and the top-ranking teams will be invited to co-author the NTIRE 2026 challenge report and to submit papers with their solutions to the NTIRE 2026 workshop. Detailed descriptions are appreciated.

The factsheet, source codes/executables, trained models should be sent to **all of the NTIRE 2026 challenge organizers (Jiatong li, Jingkai Wang, Jue Gong, Zheng Chen, Kai Liu, Yulun Zhang, and Radu Timofte)** by email.

2. Email final submission guide

To: jiatong.li2024@gmail.com
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cc: your_team_members

Title: NTIRE 2026 Mobile Real-World Image Super-Resolution Challenge - TEAM_NAME - TEAM_ID

To get your TEAM_ID, please register at [Google Sheet](#). Please fill in your Team Name, Contact Person, and Contact Email in the first empty row from the top of the sheet. Body contents should include:

- team name
- team leader's name and email address
- rest of the team members
- user names on NTIRE 2026 CodaBench competitions
- Code, pre-trained model, and factsheet download command, e.g. `git clone ..., wget ...`
- Result download command, e.g. `wget ...`
 - Please provide different URLs in e) and f)

Factsheet must be a compiled pdf file together with a zip with .tex factsheet source files. Please provide a detailed explanation.

3. Code Submission

The code and trained models should be organized according to the [GitHub repository](#). This code repository provides the basis for comparing the various methods in the challenge. **Code scripts based on other repositories will not be accepted.** Specifically, you should follow the steps below.

- Git clone [the repository](#)
- Put your model script under the `models/[Team_ID]-[Model.Name]` folder
- Put your pretrained model under the `model_zoo/[Team_ID]-[Model.Name]` folder
- Modify `model_path` in `test.py`. Modify the imported models
- `python test.py` (restore images, details in GitHub)
- `python eval.py` (eval results, details in GitHub)

Although AdcSR[2] recovers part of the performance through adversarial distillation, it is challenging to learn high-quality deep representations purely relying on diffusion objectives. To further break through the performance bottleneck of AdcSR[2], we introduce the REPA[9] Loss in the second stage of training, i.e., the distillation stage. For a given input image $\vec{X}_{LR}$, GT image $\vec{X}_{GT}$, and model hidden state h_t , we calculate the projection output $h_\phi(h_t)$ of z_t . REPA[9] aligns the projection output h_t with the deep representation result $h_\varepsilon = \mathcal{E}(\vec{X}_{GT})$ encoded by the external encoder $\mathcal{E}(\cdot)$. The REPA[9] Loss is expressed as:

$$\mathcal{L}^{(\text{repa})} = -\mathbb{E}_{\mathbf{x}_{\text{GT}}, t} \left[\frac{1}{N} \sum_{n=1}^N \text{sim}(\mathbf{h}_{\mathcal{E}}[n], h_{\phi}(\mathbf{h}_t[n])) \right] \quad (1)$$

where N is the number of feature patches, and $(\cdot)$ is the cosine similarity function. It is worth emphasizing that the projection head $h_{\phi}(\cdot)$ and the external encoder $\mathcal{E}(\cdot)$ only participate in gradient backpropagation during the training phase and are completely discarded during test inference. Therefore, we achieve quality improvement without increasing any additional inference time or parameters. In practice, we insert an MLP as $h_{\phi}(\cdot)$ after the first downsampling block in the UNet, and the pre-trained feature encoder DINOv2[3, 4, 6] is adopted as $\mathcal{E}(\cdot)$.

Combining the distillation framework of AdcSR[2] with the newly introduced representation alignment regularization, the total training loss function of our model is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{distill}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}} + \lambda_{\text{REPA}} \mathcal{L}_{\text{REPA}} \quad (2)$$

where $\mathcal{L}_{\text{distill}}$ constrains the L1 distance between the student model and the teacher model in the VAE Decoder feature space; $\mathcal{L}_{\text{adv}}$ is the non-saturating adversarial loss to improve the realism of generated images; $\mathcal{L}_{\text{REPA}}$ is the introduced representation alignment loss, and λ_{REPA} is set to 0.5 according to conventional practices.

Training Strategy Following AdcSR[2], our basic training configuration is roughly the same as the second-stage training of AdcSR[2], i.e., experiments are conducted on the Real-ISR task with a scaling factor of 4. The sizes of LR and HR images are set to 128×128 and 512×512 by default. We jointly fine-tune the UNet with 25% channel pruning and all first blocks of the pre-trained VAE decoder, where $\lambda_{\text{adv}} = 1$ and $\lambda_{\text{REPA}} = 0.5$, for 100K steps. Two datasets provided are used for training: DIV2K[1] and LSDIR[5]. We continued training on the official weights of AdcSR and set the initial learning rate to $5e - 5$. The learning rate of the discriminator and LoRA rank are set to $1e - 6$ and 4, respectively. The high-order degradation pipeline of RealESRGAN[7] is used to synthesize LR-HR image pairs. Training is performed on 8 NVIDIA Tesla V100 (32GB) GPUs with the Ada optimizer and a batch size of 8.

5. Other details

- Planned submission of a solution(s) description paper at NTIRE 2026 workshop.
- General comments and impressions of the NTIRE 2026 challenge.
- What do you expect from a new challenge in image restoration, enhancement and manipulation?

- Other comments: encountered difficulties, fairness of the challenge, proposed subcategories, proposed evaluation method(s), etc.

References

- [1] Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2017. 3
- [2] Bin Chen, Gehui Li, Rongyuan Wu, Xindong Zhang, Jie Chen, Jian Zhang, and Lei Zhang. Adversarial diffusion compression for real-world image super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2, 3
- [3] Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers, 2023. 2, 3
- [4] Cijo Jose, Théo Moutakanni, Dahyun Kang, Federico Baldassarre, Timothée Darcet, Hu Xu, Daniel Li, Marc Szafraniec, Michaël Ramamonjisoa, Maxime Oquab, Oriane Siméoni, Huy V. Vo, Patrick Labatut, and Piotr Bojanowski. Dinov2 meets text: A unified framework for image- and pixel-level vision-language alignment, 2024. 2, 3
- [5] Yawei Li, Kai Zhang, Jingyun Liang, Jiezhang Cao, Ce Liu, Rui Gong, Yulun Zhang, Hao Tang, Yun Liu, Denis Demandolx, Rakesh Ranjan, Radu Timofte, and Luc Van Gool. Lsdire: A large scale dataset for image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1775–1787, 2023. 3
- [6] Maxime Oquab, Timothée Darcet, Theo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Russell Howes, Po-Yao Huang, Hu Xu, Vasu Sharma, Shang-Wen Li, Wojciech Galuba, Mike Rabbat, Mido Assran, Nicolas Ballas, Gabriel Synnaeve, Ishan Misra, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision, 2023. 2, 3
- [7] Xintao Wang, Liangbin Xie, Chao Dong, and Ying Shan. Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In *International Conference on Computer Vision Workshops (ICCVW)*. 3
- [8] Rongyuan Wu, Lingchen Sun, Zhiyuan Ma, and Lei Zhang. One-step effective diffusion network for real-world image super-resolution. *arXiv preprint arXiv:2406.08177*, 2024. 2
- [9] Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. In *International Conference on Learning Representations*, 2025. 2